CarbonLattice Technologies — Electrical, Instrumentation & Controls

CMBSG Hardware Implementation and Development Process Guide

VOLUME 3

Electrical, Instrumentation & Control System Implementation

Reference Design Basis: 10,000 Nm^3/hr CMBSG Module
Document Class: Engineering Deliverable — E&I / Controls

Read Before Use

This volume assumes Volume 2 mechanical completion has been certified. It covers electrical distribution, instrumentation, control-system architecture, and control narratives for normal operation. **Safety Instrumented Function (SIF) logic is specified functionally in this volume for wiring and configuration purposes, but final SIL verification, proof-test intervals, and bypass management are governed by Volume 5 (Safety Systems, HAZOP Closeout & PSM) — do not commission any SIF from this volume alone.** Where a control loop's design basis was corrected in Volume 1, this volume flags it and wires/configures against the corrected value.

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1 How to Use This Volume

1.1 Scope and Boundary with Other Volumes

This volume covers: electrical power distribution to all tagged equipment, the full instrument index, control-loop wiring and logic narratives, the DCS/PLC architecture, HMI design standards, network/cybersecurity architecture, and pre-commissioning loop-check procedures. It stops short of energizing any Safety Instrumented Function for live process control (Volume 5) and stops short of introducing process fluids (Volume 4, Commissioning).

1.2 Tag Numbering Convention

This volume follows ISA-5.1 style tag numbering already established in the reference documents: first letter(s) indicate the measured/initiating variable (F=flow, T=temperature, A=analytical, L=level, P=pressure, C=conductivity), second letter indicates function (I=indicator, T=transmitter, C=controller, IT=indicating transmitter), followed by a stage-prefixed loop number (1xx=Stage 1, 2xx=Stage 2, 3xx=Stage 3, 4xx=Stage 4).

2 Electrical System Design

2.1 Load List and Single-Line Summary

Table 1: Master electrical load list, corrected basis.

Tag	Description	Rating	Voltage	Starter Type
F-101	ID fan motor	35 kW	415V, 3ph	VFD, 40–100% turndown
P-101	Sump recirculation pump	7.5 kW	415V, 3ph	DOL with soft-start
R-201 agitator	Leaching reactor agitator	11 kW	415V, 3ph	VFD, 20–80 RPM range
P-202	Stage 2 recirculation pump	15 kW	415V, 3ph	VFD
US-301/2/3	Ultrasonic transducer drivers	3×1.5 kW	415V or dedicated driver supply	Per vendor driver unit
DS-301	Enzyme dosing skid (pump + chiller)	2 kW	230V, 1ph	DOL
PM-401	Pug-mill twin-shaft drive	18.5 kW	415V, 3ph	VFD
HC-401	Hydrocyclone feed pump	5.5 kW	415V, 3ph	DOL
HP-401	Hydraulic power unit (HPU)	45 kW	415V, 3ph	DOL with star-delta or soft-start per vendor
CT-401	Curing tunnel heaters + fans	22 kW	415V, 3ph	Contactor-controlled, zoned
DCS/ instrumentation	Control room + field instruments	8 kW	230V, 1ph (UPS-backed)	UPS with min. 30 min autonomy
Total connected load		**~174 kW**		

Cross-Check Against Original Reference

The original technical reference states a continuous electrical load of 125 kW (45 kW press, 35 kW ID fan implied, remainder distributed). This volume's bottom-up load list totals approximately 174 kW connected load, which is expected — **connected load always exceeds actual continuous operating load** because of load diversity (not all VFD-driven equipment runs at 100% simultaneously) and starting/inrush allowances. Size the transformer and main incomer for the connected load with diversity factor 0.75–0.85 applied (typical for this equipment mix), giving an expected continuous demand of 130–148 kW, consistent with the original 125 kW figure within normal engineering margin.

2.2 Area Classification

Table 2: Electrical area classification by zone.

Area	Classification	Basis
Stage 1 (gas conditioning)	Unclassified (general purpose)	Aqueous/gas process, no flammable atmosphere expected under normal operation
Stage 2 (matrix blending)	Unclassified, but corrosion-rated enclosures	Acidic mist potential, not flammable
Stage 3 (mesh absorber)	Unclassified	No flammable service
HF handling area (Ti mesh prep, per Volume 2 Section 5.3 vendor process)	Unclassified but **corrosion-severe**	HF vapor is not flammable but is highly corrosive to standard electrical enclosures; specify HF-resistant enclosure coatings
$NaBH_3CN$ / HCN handling area	Unclassified but **toxic gas monitoring required**	See Volume 5; electrical equipment in this area should be rated for the ambient corrosive/toxic exposure per equipment vendor's chemical resistance data
Stage 4 hydraulic press	Unclassified	Hydraulic oil present but not classified as a flammable atmosphere source at normal operating temperature

Note — No CBRN or Explosive Atmosphere Classification Required

Nothing in the corrected process basis (Volume 1) introduces a flammable or explosive atmosphere requiring Zone 1/2 or Class I Div. 1/2 electrical classification. The principal electrical-design driver in this plant is **corrosion resistance** (acidic mist, HF vapor) and **washdown/ingress protection**, not explosion-proofing. Confirm this assumption against your specific site's fire and gas study during Volume 5 HAZOP closeout before finalizing enclosure ratings.

2.3 Grounding and Bonding

Grounding Requirements Summary

- Equipment grounding conductor sized per local electrical code (e.g., IS 3043 or IEC 60364 depending on jurisdiction) for the largest connected load on each circuit.
- **Dissimilar-metal bonding care**: Super Duplex (T-101), 316L SS (T-202, C-301 shell), Hastelloy C-276 (P-202), and titanium (C-301 mesh) equipment must each have an independent bonding jumper to the plant grounding grid — do not rely on pipe-to-pipe continuity across dielectric isolation unions (Volume 2, Section 7) as a grounding path, since those unions are specifically installed to break galvanic continuity.
- Instrumentation grounding: single-point ground for all 4-20mA loop shields at the marshalling cabinet only (not at the field instrument end) to avoid ground loops.
- Static grounding: chitosan solution transfer lines (Stage 2) and titanium mesh handling fixtures (Stage 3, per Volume 2 Section 5.3) should have static-dissipative grounding straps during maintenance access, consistent with general good practice for particulate/mist-handling equipment.

3 Master Instrument Index

3.1 Stage 1 Instruments

Table 3: Stage 1 instrument index.

Tag	Service	Range	Signal / Type
FIT-101	Flue gas flow	0–12,000 Nm^3/hr	4-20mA, thermal mass or pitot-based flow element
TIT-101	Gas inlet temperature	0–250 °C	4-20mA, RTD (Pt100)
AIT-101	Inlet SO_2 analyzer	0–2,500 mg/Nm^3	4-20mA, UV-fluorescence or NDIR analyzer
LIT-101	Sump level	0–100%	4-20mA, radar or guided-wave
AIT-102	Sump pH	0–14 pH	4-20mA, glass electrode, redundant pair (SIS-adjacent)
PIT-102	Throat differential pressure	0–10 kPa	4-20mA, differential pressure transmitter; **SIS-A input, see Volume 5**
SIT-101	Conductivity	0–10,000 μS/cm	4-20mA, inductive conductivity probe

3.2 Stage 2 Instruments (Including New SI Trim Loop)

Table 4: Stage 2 instrument index.

Tag	Service	Range	Signal / Type
TIT-201	R-201 reactor temperature	0–80 °C	4-20mA, RTD
AIT-201/202	R-201/T-202 pH	0–14 pH	4-20mA, redundant pair, chemical-resistant electrode (aggressive leachate service)
LIT-201/202	R-201/T-202 level	0–100%	4-20mA, radar
FIT-201	Slag feed rate	0–400 kg/hr	Loss-in-weight feeder signal, 4-20mA
AIC-202-TRIM	**SI trim sub-loop controller (new per Volume 1/2 correction)**	**0–5 L/hr dosing**	**4-20mA output to trim metering pump, cascaded from pH/conductivity soft-sensor — see Section 7.3**
CIT-202	T-202 conductivity (SI soft-sensor input)	0–20,000 μS/cm	4-20mA

3.3 Stage 3 Instruments

Table 5: Stage 3 instrument index.

Tag	Service	Range	Signal / Type
AIT-301	Inlet CO_2	0–20 vol%	4-20mA, NDIR analyzer
AIT-302	Outlet CO_2	0–5 vol%	4-20mA, NDIR analyzer; **primary performance-verification instrument for the Hatta-number open item, see Section 7.4**
AIT-303	Outlet SO_2	0–50 mg/Nm^3	4-20mA
TIT-301	Tower mid-bed temperature	0–80 °C	4-20mA, RTD
CIT-301	Carbonic anhydrase activity	0–15,000 WAU/mg	Lab-verified with soft-sensor correlation (see Volume 1 Section 5.4 note on soft-sensor use)
FIT-301	Matrix flow rate	0–5 L/s	4-20mA, magnetic flowmeter
PIT-301	Tower differential pressure	0–3 kPa	4-20mA; **SIS-C input, see Volume 5**
LIT-301	Sump level	0–100%	4-20mA, radar

3.4 Stage 4 Instruments

Table 6: Stage 4 instrument index.

Tag	Service	Range	Signal / Type
PIT-401	Hydraulic system pressure	0–250 bar	4-20mA, pressure transmitter; relief valve set 220 bar (Volume 1)
ZIT-401	Press ram position	0–300mm	4-20mA, linear position transducer
TIC-401	Curing tunnel temperature control	0–80 °C	4-20mA, RTD, PID control to 55–60 °C setpoint
MIT-401	Curing tunnel humidity	0–100% RH	4-20mA, capacitive RH sensor rated for the tunnel's warm/humid environment
WIT-401	Block weight check	0–5 kg	Load cell, 4-20mA, in-line QC per Volume 2/6

3.5 HSE/Safety-Related Instruments (Wiring Only — Logic in Volume 5)

Table 7: HSE-related instrument index (functional wiring; SIL logic per Volume 5).

Tag	Service	Range	Signal / Type
AIT-401/402	Reduction bath pH ($NaBH_3CN$/HCN risk, redundant pair)	0–14 pH	4-20mA, redundant 2-out-of-2 voting per Volume 5
GIT-401	HCN gas monitor	0–50 ppm	4-20mA, electrochemical or photoionization detector
CIT-401	HF drip-tray conductivity (spill detection)	0–5,000 μS/cm	4-20mA

4 Control System Architecture

4.1 DCS/PLC Selection Criteria

Table 8: Control system architecture summary.

Element	Specification
Primary controller	Redundant PLC pair (hot-standby) or small-scale DCS, minimum 20% spare I/O capacity for future expansion (e.g., the tower extension section flagged in Volume 2, Section 5.3)
Safety controller	**Physically and logically separate** Safety Instrumented System (SIS) controller, SIL-2 rated minimum, independent of the basic process control system (BPCS) — do not implement SIF logic in the same controller as normal process control
I/O count (estimated)	Approximately 45–55 analog I/O, 30–40 digital I/O across all four stages plus HSE instruments; confirm exact count against Section 2 instrument index plus motor control interlocks
HMI	Redundant server pair, minimum 3 operator workstations (control room, plus mobile/tablet access for field verification per Volume 6 SOPs)
Historian	Minimum 1-year data retention at 1-second scan rate for critical loops (AIT-302 outlet CO_2, PIT-401 hydraulic pressure, AIT-401/402 reduction bath pH)

4.2 Network Topology and Redundancy

Network Architecture Summary

- Redundant ring or star-ring hybrid fieldbus network (e.g., PROFINET, Foundation Fieldbus, or vendor-standard protocol) connecting field marshalling cabinets to the controller room.
- Separate, air-gapped or firewall-segmented network for the SIS controller (Section 3.1) — SIS network must not share a switch or cable path with the BPCS network beyond a single, monitored, unidirectional data-diode connection for historian purposes only.
- Enzyme dosing skid (DS-301) and HSE gas monitors (GIT-401, CIT-401) on a dedicated, higher-priority polling group given their role in the pending pilot-verification and safety-critical monitoring respectively.

4.3 Cybersecurity Segmentation

Table 9: Network security zones (per ISA/IEC 62443 zone-and-conduit model).

Zone	Contents	Conduit Rules
Zone 0 (Safety)	SIS controller, safety I/O	No inbound connections except engineering workstation via physical keyswitch enable
Zone 1 (Control)	BPCS controller, field I/O	Firewalled from Zone 2; historian data-diode only to Zone 0
Zone 2 (Supervisory)	HMI servers, historian, engineering workstation	Firewalled from corporate IT (Zone 3); no direct internet access
Zone 3 (Enterprise)	Corporate IT, remote access VPN	Remote access to Zone 2 only via jump-host with MFA, logged and time-limited

5 Control Narratives

5.1 Stage 1 Control Narrative

Stage 1 Normal Control Philosophy

- ID fan (F-101) speed is modulated via VFD to maintain a slight negative draft at the Stage 1 inlet, cascaded from a duct pressure setpoint (not separately tagged above but implied by the fan control loop — add PIT-100 if not already present in the site-specific P&ID).
- Quench water flow is ratio-controlled to gas flow (FIT-101) to maintain the 45 °C saturated outlet target; TIT-101 outlet reading provides trim correction.
- Sump level (LIT-101) controls the ash slurry bleed-off rate to Stage 4, targeting the 40 wt% solids design point.
- AIT-102 (sump pH) is a monitored/alarmed variable in Stage 1 — there is no active pH control loop here (the acidic condensate is a natural byproduct, not a controlled setpoint), but a low-pH alarm at pH 2.0 alerts operators to unusual SO_3 carryover per the original HAZOP finding.

5.2 Stage 2 Control Narrative — Including the SI Trim Loop

Corrected Control Philosophy for T-202

The original reference's design assumed a single acid-dosing point (at R-201) would maintain SI < 0 throughout Stage 2. Volume 1's finding that SI recomputes to $+0.42$ from the paper's own numbers means **this volume specifies cascade control, not single-loop control, for T-202 chemistry**:

1. Primary loop: AIT-202 (T-202 pH) controls the main acid dosing valve at R-201 outlet, slow-responding (reactor-scale time constant).
2. **New secondary/trim loop**: AIC-202-TRIM receives a computed SI soft-sensor value (from AIT-202 pH and CIT-202 conductivity, using the ion-balance approach in Volume 1 Section 4.2) and trims the fast-acting metering pump (installed per Volume 2, Section 4.4) to hold SI < -0.3 continuously, correcting on a much faster time constant than the primary loop.
3. Alarm: SI soft-sensor computed value > -0.1 triggers a high-priority operator alarm (not a full SIS trip — this is a process-quality issue, not an immediate hazard, per Volume 5's HAZOP classification), prompting operator verification and manual acid addition if the trim loop cannot recover the trend within 10 minutes.

5.3 Stage 3 Control Narrative — Enzyme Dosing Under the Pending Pilot Verification

Provisional Control Philosophy — Subject to Volume 4 Pilot Result

Given the Hatta-number finding (Volume 1, Section 5.1), the enzyme dosing control loop is specified here as **adaptive rather than fixed-setpoint** until Volume 4's pilot verification confirms actual capture-efficiency performance:

1. DS-301 dosing rate has an operator-adjustable range of 5–20 mg/L (wider than the original single-point 12 mg/L design target) specifically to allow the commissioning team to run the sensitivity trial referenced in Volume 1.
2. AIT-302 (outlet CO_2) is the primary performance-verification instrument. Configure a trend/report function comparing AIT-301 (inlet) vs. AIT-302 (outlet) in real time so the capture efficiency $\eta_{CO_2} = 1 - \text{AIT-302}/\text{AIT-301}$ is continuously visible on the HMI — this is the actual, empirical answer to the open Hatta-number question, and should be trusted over the original design correlation until reconciled.
3. Do not hard-code the original reference's capture-efficiency formula (Section 11.5 of the technical reference, $\eta_{CO_2} = 1 - \exp(-k_{CA} \cdot \tau_{res} \cdot 12000)$) into any control or optimization logic without first validating k_{CA} against commissioning data — this formula's underlying enhancement-factor assumption is exactly what Volume 1 flagged as unverified.

5.4 Stage 4 Control Narrative

Stage 4 Normal Control Philosophy

- PM-401 pug-mill speed is fixed-speed (or narrow VFD trim range) per vendor recommendation; not a tight control loop.
- HP-401 press cycle is sequence-controlled (not continuous PID): fill → compress to 200 bar → 30s dwell → release → eject, triggered by ZIT-401 position feedback and PIT-401 pressure feedback.
- CT-401 curing tunnel: TIC-401 and MIT-401 (humidity) are both PID-controlled to the 55–60 °C / 90% RH setpoints. **Per Volume 2's Arrhenius rate-constant correction, configure the tunnel control system with an adjustable residence-time setpoint (8–14 hours) rather than hard-coding 12 hours**, to support the cure-time sensitivity trial recommended for commissioning (Volume 4).

6 HMI Screen Design Standards

Table 10: HMI screen hierarchy and design standards.

Screen Level	Content
Level 1 — Overview	Full plant PFD-style mimic (per original reference Appendix B), color-coded by stage, with headline KPIs: CO_2 capture efficiency (real-time, per Section 7.4 above), block production rate, current alarm summary
Level 2 — Stage detail	One screen per stage, full instrument list with live values, trend faceplates, and manual/auto control station access
Level 3 — Loop faceplate	Individual PID faceplate: PV, SP, OP, mode, tuning parameters (engineer access only)
Level 4 — Diagnostics	Instrument diagnostics, communication health, historian query tools
Alarm philosophy	Follow ISA-18.2 alarm management standard: prioritize (critical/high/medium/low), avoid alarm flooding, and specifically **do not alarm-flood on the AIC-202-TRIM loop** during initial commissioning while its tuning is being established — use a temporary suppressed/shelved state per ISA-18.2 procedure instead of disabling the alarm outright

Color Convention

Reuse the color convention established in the technical reference documents throughout this project: emerald green for CO_2/carbon-related tags, sky blue for water/cooling tags, amber for corrections/attention items, rose for alarms/warnings. This consistency reduces operator training time (Volume 8) since the visual language matches the engineering documentation the operators will also reference.

7 Cable Schedule and Marshalling

Table 11: Cable schedule summary by category.

Cable Category	Type	Routing Note
4-20mA instrument signal	Individually shielded twisted pair, 2-core	Separate tray from power cable, minimum 300mm separation
Power (motors >5kW)	Armored, per motor rating	Dedicated tray, VFD output cables shielded to limit EMI onto instrument tray
Fieldbus/network	Per protocol spec (e.g., shielded twisted pair or fiber)	Fiber preferred for runs >100m or through high-EMI areas (near VFDs)
Safety (SIS) wiring	Dedicated, separately identified (e.g., red jacket)	Physically separate tray/conduit from BPCS wiring per Volume 5

8 Loop Check and Pre-Commissioning Test Procedures

Standard Loop Check Procedure

1. Verify field instrument tag, range, and calibration certificate match the instrument index (Section 2) before wiring.
2. Simulate 0%, 50%, 100% signal at the field instrument terminal (or via HART communicator for smart transmitters); confirm correct reading at the DCS/PLC and correct engineering unit scaling.
3. Verify alarm setpoints are configured per the instrument index and (for SIS-related tags) per Volume 5's SIF specification — **do not enable any SIS action during this loop check; verify indication only, actuation testing is a separate Volume 5 procedure**.
4. For control loops (e.g., TIC-401, AIC-202-TRIM), verify output signal reaches the correct final control element and that direction of action (direct/reverse) is correct — an inverted control loop can cause a runaway condition on first automatic-mode transfer.
5. Document loop check completion on a per-tag basis; all loop checks must be complete and signed off before Volume 4 commissioning begins.

A Appendix A: Blank I/O List Template

Tag	Description	Signal Type	Range	I/O Slot	Card	Calibration Due

Table 12: Blank I/O list template for use during detailed engineering.

B Appendix B: Loop Diagram Template

Field	Entry
Loop tag	
Field instrument model/manufacturer	
Wiring terminal numbers (field, junction box, marshalling, controller)	
Power supply source	
Range and units	
Alarm/trip setpoints	
Associated SIF (if any, cross-reference Volume 5)	

Table 13: Blank loop diagram template — complete one per instrument loop.

C Appendix C: Cross-Reference to Volume 1/2 Corrections

Prior Volume Finding	E&I / Control Impact	Section
Saturation index sign error (Vol. 1)	New AIC-202-TRIM loop, cascade control philosophy	Sections 2.2, 5.2
Hatta number / enhancement factor unsupported (Vol. 1)	Adaptive enzyme dosing range 5–20 mg/L, real-time capture-efficiency HMI display, do not hard-code original formula	Section 5.3
Arrhenius rate constant error (Vol. 1)	Adjustable cure-time setpoint 8–14 hr instead of fixed 12 hr	Section 5.4

Prior Volume Finding	E&I / Control Impact	Section
Tower bolt-on extension provision (Vol. 2)	I/O and controller spare capacity (20%) reserved for additional tier instrumentation if tower is extended	Section 3.1
Stage 1 ΔP off by 10× (Vol. 1)	PIT-102 range specified 0–10 kPa with sufficient margin either way; SIS-A setpoint unchanged at 5 kPa	Table 3